VIGHNESH INC.

ML RESEARCH ENGINEER / SCIENTIST

TAKE-HOME TECHNICAL ASSESSMENT

Topic: Temporal & Multimodal Visual Intelligence

Submission Window: 7 Days
Submission: Git repository + short technical report

1. Objective

Build a small experimental system to investigate the following question:
Can a model extract and utilize useful information from a sequence of visual observations that may not be available from an individual frame?

The assessment is intentionally open-ended. We are interested not only in the final performance, but in **how you formulate the problem, design the experiment, select the model, validate your findings, and reason about limitations.**

You may use publicly available datasets and pretrained models.

2. Dataset & Sequence Analysis

Select a suitable **video / egocentric / sequential vision dataset** containing ordered observations.
You may use a small subset; downloading a massive dataset is unnecessary.
Your pipeline should:

- Load ordered sequences.
- Verify frame ordering and sequence integrity.
- Generate basic dataset statistics.
- Visualize several sequences as temporal frame strips.
- Identify examples where temporal information appears potentially useful.

Briefly discuss:

- What changes across the sequence?
- When does additional information become available?
- Are there examples where a single frame is ambiguous?
- Are there examples where temporal information appears unnecessary?

3. Visual Representation / Embeddings

Use a pretrained visual representation model such as:

- DINOv2
- CLIP / SigLIP
- another suitable pretrained vision encoder

Extract an embedding for each frame.
For example:
Frame 1 → E1
Frame 2 → E2
Frame 3 → E3
...
Perform a basic analysis of how representations change over time.
At minimum, investigate one or more of:

- cosine similarity between consecutive frames;
- similarity between different sequences;
- temporal embedding trajectory;
- simple temporal pooling.

Include **at least one visualization** and briefly explain your observations.

4. Single-Frame vs Temporal Model
Implement two approaches.
Baseline
A model that makes its prediction using **one frame / one visual representation**.
Temporal
A model that receives a sequence of frame representations.
You may use a:

- GRU / LSTM
- Transformer Encoder
- Temporal CNN
- other justified temporal architecture

Keep the model reasonably small.
The objective is not to build a state-of-the-art architecture.
It is to investigate whether **sequence information changes the result.**

5. Temporal Evaluation
Perform at least **two controlled experiments** from the following:
Progressive Observation
Evaluate using:
1 frame
2 frames
3 frames
...
N frames
and observe how prediction/performance changes.
Temporal Order
Compare:
Correct sequence
against:
Shuffled sequence
Occlusion / Corruption
Artificially degrade selected observations through techniques such as:

- blur
- masking
- dropped frames
- noise
- partial occlusion

Compare the single-frame and temporal approaches.
Viewpoint / Appearance Variation
Where possible, evaluate performance across different viewpoints, environments or appearances.

6. Multimodal / VLM Component
Add a small experiment involving a **Vision-Language Model (VLM)**.
The goal is not to fine-tune a VLM.
Use a suitable VLM/API/open-source model to perform one or more of:

- sequence-level description;
- temporal change analysis;
- semantic labeling;
- identification of ambiguous observations;
- comparison between early and later observations.

For example:
Given the sequence, describe what changes between observations and what additional information becomes available over time.
Briefly discuss:
Where does the VLM add value compared with the visual embedding/model?

7. Research Analysis

This section is more important than achieving a particular accuracy.

Answer concisely:

A.

When did temporal information help?

B.

When did it not help?

C.

How can you determine whether a temporal model is actually using temporal relationships rather than simply receiving more visual information?

D.

What experimental result would convince you that the temporal approach is **not useful for the selected task**?

E.

What would you change in the dataset or experiment to obtain stronger evidence?

8. System Design

Provide **one architecture diagram** showing how you would evolve your experiment into a larger multimodal perception system.

It should conceptually address some/all of:

Visual Input

↓

Visual Representation

↓

Temporal Modeling

↓

Multimodal Information

↓

Semantic / VLM Layer

↓

Prediction / Understanding

You may introduce additional modalities such as:

- Depth
- IMU
- Pose
- audio
- spatial information

if you believe they are relevant.

Briefly explain your design choices.

9. Engineering Requirements

Your submission should contain:

README.md

src/

  data/

  models/

  embeddings/

  evaluation/

  vlm/

scripts/

results/

report.pdf

requirements.txt

The exact structure is up to you.

The README should allow another engineer to understand:

1. Environment setup
2. Dataset preparation
3. Embedding extraction
4. Training
5. Evaluation
6. Reproduction of your results

Clean, understandable and reproducible implementation is expected.

10. Deliverables

1. Source Code

Complete runnable implementation.

2. Technical Report

Approximately **4–6 pages**, covering:

- Dataset
- Approach
- Architecture
- Experiments
- Results
- Analysis
- Limitations
- Future improvements

3. Results

Include appropriate:

- metrics
- plots
- sequence visualizations
- comparison between approaches
- failure cases

4. Short Demo

Optional but preferred:

5-minute screen recording demonstrating the system and key findings.

11. What We Evaluate

There is **no predetermined target accuracy**.

We will primarily evaluate:

Area	**Importance**
Problem understanding	High
Experimental design	Very High
Temporal reasoning	Very High
ML fundamentals	High
Embedding understanding	High
Transformer / sequence modeling	High
VLM understanding	Medium–High
Data & evaluation discipline	Very High
Code quality	High
Research thinking	Very High

12. Important
Do not optimize solely for a high metric.
A technically honest conclusion such as:
"The temporal model did not improve this dataset because the task can largely be solved from individual observations."
can be more valuable than an unexplained 95% accuracy.
We are interested in **how you think, experiment, validate and communicate**, not simply the final number.

Bonus — Optional
If you have additional time, you may explore one of:

- multimodal fusion;
- temporal attention visualization;
- contrastive temporal learning;
- future-state prediction;
- lightweight/edge inference;
- self-supervised temporal learning.

Bonus work is not required.

Submission
Please submit:
Git repository + PDF report + I/O Screenshots + Additional Stuffs (if any required/used)
within **7-9 days of receiving this assessment**.

Guidelines & NDA

By accepting this assessment, you agree to the following:

- **Confidentiality:** The assessment, datasets, code, documentation, and any related information are confidential. Do not share, reproduce, publish, or distribute them in any form.
- **Independent Work:** The submission must be your own work. Collaboration, outsourcing, plagiarism, or submitting another person's work is not permitted.
- **Third-Party Resources:** Public documentation, libraries, papers, and standard development tools may be used where appropriate. Clearly acknowledge substantial third-party code, data, or other resources.
- **Security & Privacy:** Do not attempt to access systems, credentials, data, repositories, or resources outside the scope of the assessment. Do not include confidential information belonging to your current or previous employer.
- **Submission:** Please submit only the materials required for evaluation and remove any secrets, API keys, passwords, tokens, or unrelated confidential information.
- **Intellectual Property:** Your pre-existing IP, general knowledge, skills, and independently developed materials remain yours. The submitted assessment may be used by **Vighnesh Inc.** solely for evaluating your candidacy, unless otherwise agreed in writing.
- **Evaluation:** The assessment will be evaluated based on technical reasoning, implementation quality, research methodology, experimentation, results, and clarity of communication. Alternative approaches are welcome when properly justified.
- **Deadline:** Please submit your assessment within 7-9 days of receiving this assessment.